COMMERCIAL SOLUTION BRIEF | ADVANCED AQUACULTURE TECHNOLOGIES

aquaCON™ Intelligent Solar Feeder System

Autonomous Precision Dosing, Long-Range LoRaWAN® Mesh & Real-Time Cloud Intelligence

15% – 22%	100% Solar	15 km Range	₹0 / Feeder	< 1 Crop
FCR Improvement	**Zero Shore Cabling**	**LoRaWAN® Telemetry**	**Zero Feeder SIMs**	**Full Payback**
(Saves 8–10 Tons Feed)	(LiFePO4 Powerplant)	(Farm-Wide Coverage)	(1 Central Gateway)	(ROI in 90–120 Days)

The Industry Challenge: Costly Feeding Inefficiency

Feed accounts for **50% to 70% of total operational expenditure** in commercial aquaculture. Traditional manual hand-feeding and dumb timer dispensers suffer from crippling flaws:

- **Feed Wastage & High FCR:** Hand-broadcast feeding is irregular. Uneaten feed accumulates on pond bottoms, rotting into toxic ammonia spikes and causing severe shrimp mortality.
- **Blind Conventional Dispensers:** Mechanical timers dump feed blindly during hopper jams or empty feed situations, burning out motors and leaving shrimp starved.
- **Dangerous Underwater Cabling:** Running 230V AC shore power across saline water creates severe electrocution hazards, frequent tripping, and expensive pond trenching.
- **High Recurring Telecom Costs:** Installing individual cellular SIM cards on 50+ feeders wastes over **₹2,00,000+ per year** in recurring monthly data bills.

The aquaCON™ Solution: Autonomous Precision Dosing

aquaCON™ is an industrial-grade automated feeding system engineered specifically to slash Feed Conversion Ratio (FCR) and maximize farm profitability:

- **Dynamic Schedule Recalculation:** If a pause occurs or the hopper is refilled mid-day, the smart edge processor recalculates remaining intervals to dispense the exact target quota before dusk.
- **Optical & Inductive Jam Interlocks:** Rotary stroke sensors verify full paddle discharge; automatic reverse-clearance pulses eliminate feed chute bridging.
- **Acoustic Parabolic Distribution:** High-velocity precision impeller broadcasts pellets smoothly in a calibrated 8–15 m arc with zero crumble.
- **100% Solar Power Autonomy:** Integrated 50W monocrystalline solar PV and automotive-grade LiFePO4 battery deliver 7+ days of sunless operation.



Figure 1: aquaCON™ Solar Smart Feeder broadcasting calibrated feed pellets across pond water surface at dawn with long-range LoRaWAN® RF telemetry link.

Key Engineering Innovations:

► Sub-Gram Dosing Precision:
Calibrated micro-dosing (50 g to 500 g per burst) with ±1.5% repeatability ensures shrimp consume feed instantaneously without nutrient leaching.

► Anti-Bridging Chute Geometry:
Specialized low-friction hopper cone with integrated agitator prevents pellet compaction across all pellet sizes (0.8 mm to 3.5 mm).

► Long-Range LoRaWAN® Telemetry:
Transmits dispensed kilograms, battery health, solar charging, and motor status up to 15 km line-of-sight to the central farm gateway.

► Heavy-Duty Marine Ingress Protection:
UV-stabilized virgin polymer hopper and 316 stainless steel chassis engineered for corrosive coastal saline environments.

aquaCON™ Feeder Hardware Specifications

Hopper Capacity:	120 kg / 250 kg UV Virgin Polymer	**Dosing Precision:**	50 g – 500 g per burst (±1.5%)
Spread Radius:	8 to 15 meters parabolic arc	**Supported Pellets:**	0.8 mm to 3.5 mm floating / sinking
Solar Power:	50W Monocrystalline PV Panel	**Battery System:**	12.8V LiFePO4 Smart BMS (7+ days reserve)
In-Pond Wireless:	LoRaWAN® Sub-GHz (865–868 / 915–928)	**RF Range:**	Up to 15 km to Central Farm Gateway
Ingress Rating:	IP67 Weatherproof & Marine Grade	**Jam Protection:**	Inductive Rotary Sensor + Reverse Pulse
Operational Mode:	Dynamic Schedule (06:00 to 18:00)	**Safety Interlocks:**	Motor Overcurrent, Empty Feed Detection

Enterprise Architecture & Competitive Differentiation

From Pond Edge to Executive Cloud: Complete Visibility, Zero SIM Overhead, Proven ROI

aquaCON™ Solution Brief
Architecture & Financial Model



3-Tier Scalable IoT Farm Ecosystem:

Tier 1: Intelligent Feeder Nodes (Pond-Side):
Autonomous solar feeders located on pond piers or floating frames. Each node continuously monitors battery status, solar harvesting, motor drive current, and dispense cycle completions, broadcasting lightweight telemetry packets over LoRaWAN®.

Tier 2: Central Industrial Gateway (Farm Mast):
A single high-gain industrial gateway mounted on a central mast covers **50 to 200+ ponds across a 15 km radius**. Aggregates all pond uplinks and bridges to the cloud via **tri-redundant Wi-Fi, Industrial Ethernet, or 4G LTE cellular**. Eliminates per-feeder cellular SIM costs.

Tier 3: aquaCON™ Cloud Intelligence Platform:
Enterprise cloud infrastructure provides real-time supervisor dashboards, automatic daily schedule calculation, historical event auditing, and instant SMS/call alerts for hopper empty or jam events.

Competitive Differentiation Matrix: Why aquaCON™ Outperforms

Core Capability	Manual Hand Feeding	Traditional Timer Feeders	aquaCON™ Smart Feeder
Feed Conversion Ratio (FCR)	1.65 – 1.85 (High Feed Waste)	1.45 – 1.60 (Blind Fixed Cycles)	✓ **1.20 – 1.30 (15%–22% Saved)**
Labor Deployment	4 – 6 Field Workers per 10 Ponds	2 Workers (Frequent manual checks)	✓ **1 Supervisor manages 50+ Ponds**
Power Infrastructure	Manual labor dependent	Dangerous 230V Shore Cables in Water	✓ **100% Solar + LiFePO4 (Zero Cables)**
Farm Wireless Connectivity	None (Vulnerable paper logs)	None / Short-Range Bluetooth (<15 m)	✓ **Long-Range LoRaWAN® (15 km)**
Annual SIM Card Cost	₹0 (No connectivity)	₹3,500 – ₹5,000 / feeder / year	✓ **₹0 Per Feeder (1 Central Gateway)**
Jam & Empty Interlock	Discovered hours later (Starvation)	None (Burned motors, wasted schedule)	✓ **Real-Time Sensor Interlock & Alert**
Dynamic Schedule Recalc	None (Manual human guesswork)	Impossible on mechanical timers	✓ **Autonomous Mid-Day Recalculation**
Audit Trail & Cloud App	None	None	✓ **Real-Time Cloud App & Event History**

Central Industrial Gateway & Cloud Infrastructure Specifications

Gateway Coverage:	Up to 15 km Radius Line-of-Sight	**Farm Capacity:**	Supports 50 to 200+ Feeder Nodes
Backhaul Network:	Tri-Redundant 4G LTE, Ethernet, Wi-Fi	**SIM Requirement:**	Only 1 Single Central SIM for Entire Farm
Power System:	Industrial Solar PV + Battery Backup	**Cloud Platform:**	AWS IoT Telemetry Core & DynamoDB
Security:	End-to-End AES-128 Encryption	**Alert Dispatch:**	Instant SMS / Voice Alerts for Jam / Empty

Customer ROI Model: 10-Hectare Farm (Indian Rupees)

Commercial baseline: 10 ponds consuming 40,000 kg feed per crop:

- **Feed Cost Savings:** Reducing FCR from 1.60 to 1.30 saves **7,500 kg of commercial feed** per harvest. At ₹100/kg, this delivers **₹7,50,000 to ₹10,00,000 direct cash savings per crop**.
- **Labor Payroll Savings:** Reduces feeding labor from 4 workers to 1 supervisor, saving **₹1,80,000 to ₹2,40,000 per crop cycle**.
- **Telecom SIM Savings:** Eliminates ₹2,00,000+ per year in wasted cellular data fees across farm feeders.
- **Rapid Capital Payback: Complete investment recovered in under 90 to 120 days (Within 1 crop cycle)**.

Athena Engineering Corp | Kakinada, India

Turnkey Farm Deployments & Direct Support:

- **On-Farm Pilot Deployments:** Turnkey installation of solar feeders and central gateway with on-site technician calibration.
- **Multi-Species Precision:** Calibrated for Vannamei shrimp, Black Tiger shrimp, Tilapia, and high-density marine cages.
- **Enterprise SLA & Warranty:** 3-year hardware warranty with local engineering support from Kakinada.

Schedule an On-Farm Demonstration & Site Survey

Athena Engineering Corp | Kakinada, Andhra Pradesh, India
Address: **2-48-22, 1st Street, 100 Buildings, Kakinada - 533003, AP**
Direct Commercial Lead: **+91 91546 96777** | rc@athenaenggcorp.com
Official Website: **www.athenaenggcorp.com**